

BALAJI V

Mechanical Design Engineer | FEA Specialist | SolidWorks | CSWP Certified

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PROFESSIONAL SUMMARY

Results-driven Mechanical Design Engineer with approximately one year of hands-on industry experience designing, analysing, and validating structural components for Indian Railways LHB coaches and water treatment pressure vessels. CSWP-certified (December 2025) with deep expertise in SolidWorks 3D modelling, SolidWorks Simulation (FEA), sheet metal design, weldment design, GD&T, and fabrication drawing generation. Experienced in conducting static structural FEA, mesh convergence studies, hand calculations, and design optimisation to meet structural and manufacturing requirements. Adept at creating detailed engineering documentation, BOMs, and manufacturing drawings. Passionate about applying finite element analysis and design validation to develop reliable, cost-effective mechanical components and pressure vessel systems.

WORK EXPERIENCE

Water DNA Pvt. Ltd.

2025 – Present

Mechanical Design Engineer

- Designed and developed structural sheet metal components for Indian Railways LHB coaches, including End Wall (PN 75415001), Side Wall (5-window cutout panels), 1500 L Fuel Tank, and 450 L Water Tank, using SolidWorks; produced fully annotated manufacturing drawings with GD&T callouts.
- Engineered a 3-Tier Berth System (PN LS61214 / LS61216 / LS61217) in SS 409M and MS, incorporating gas-strut-assisted upper berth mechanism and turnbuckle-adjustable middle berth; delivered complete SolidWorks assemblies and fabrication drawings.
- Designed an Air Reservoir for the LHB coach pneumatic brake system in mild steel, including saddle cradle brackets with triangular gussets; validated mounting geometry against space constraints.
- Developed water treatment pressure vessels and skid systems, including a VSL 8-6 Filter Vessel (1500×2000 mm vertical pressure vessel) and a 6000 LPH RO Plant on MS skid with multi-stage high-pressure pump and colour-coded piping layout.
- Conducted Static Structural FEA on a 25 KL SS 304 water storage tank using SolidWorks Simulation; performed mesh convergence study (refined mesh: ~3.8 M nodes / ~1.8 M elements, curvature-based) and identified peak von Mises stress of 380.9 MPa at frame-panel junctions, informing design reinforcement.
- Performed hydrostatic leak test on the fabricated 25 KL SS 304 tank (25,000-litre full-fill); verified zero leakage across all welded seam joints, nozzle connections, and panel-to-frame junctions, confirming weld integrity prior to site deployment.
- Generated comprehensive Bills of Materials (BOMs), weld symbols, tolerance stacks, and fabrication drawings for all components; maintained engineering documentation aligned with IS 2062 and ASME standards.
- Collaborated with the fabrication workshop and QC team during design reviews to resolve DFM/DFA issues, reducing rework cycles and improving first-pass fabrication quality.
- Applied reverse engineering techniques to re-model existing components from physical measurements, enabling accurate documentation of legacy parts without original drawings.

KEY PROJECTS

25 KL Water Storage Tank — Static Structural FEA — SolidWorks Simulation | SS 304

- Objective: Evaluate structural integrity of a 25,000-litre SS 304 modular storage tank under full hydrostatic load.
- Modelled full tank assembly (5 mm Base-Flange wall panels, Angle Iron 35×35×5 mm frame, C Channel 80×8 mm stiffeners; total mass ~2,855 kg) in SolidWorks and applied Static Structural FEA using SolidWorks Simulation.
- Conducted mesh convergence study; refined mesh yielded ~3.8 M nodes / ~1.8 M elements (curvature-based). Authoritative results — Static Study 7: Max von Mises Stress 380.9 MPa | Max Displacement 1.711 mm.
- Outcome: Stress concentrations identified at frame-panel junctions; analysis findings used to inform frame reinforcement decisions. Physical validation completed via hydrostatic leak test — zero leakage confirmed.

LHB Coach End Wall (PN 75415001) — SolidWorks | SS 304 | Sheet Metal

- Designed and modelled the structural end wall panel for LHB coaches in SS 304 sheet metal using SolidWorks. Generated fully dimensioned fabrication drawings with GD&T annotations and flat-pattern layouts for manufacturing.

3-Tier Berth System (PN LS61214 / LS61216 / LS61217) — SolidWorks | SS 409M + MS | Weldment

- Designed the complete 3-tier berth assembly for LHB sleeper coaches in SS 409M and mild steel. Integrated gas-strut upper berth actuation and turnbuckle middle berth adjustment. Delivered SolidWorks assemblies, weldment drawings, and fabrication BOM across 10-image documented design.

6000 LPH Reverse Osmosis Plant — SolidWorks | MS Skid | Piping Layout

- Designed a complete 6000 LPH RO plant skid in mild steel, incorporating an 80-80-40 mm three-layer membrane array, multi-stage high-pressure pump, and colour-coded piping system for inlet, permeate, concentrate, and drain lines. Produced fabrication drawings and piping layout documentation.

TECHNICAL SKILLS

CAD Software: SolidWorks (Advanced), CATIA, AutoCAD

Simulation & FEA: SolidWorks Simulation, Static Structural Analysis, Mesh Convergence Studies, Von Mises Stress Analysis, Displacement Analysis, Hand Calculations, Design Validation

Mechanical Design: Sheet Metal Design, Weldment Design, Structural Design, Pressure Vessel Design, Fabrication Design, Skid System Design, Design for Manufacturing (DFM), Design for Assembly (DFA), Reverse Engineering

Engineering Documentation: GD&T (Geometric Dimensioning & Tolerancing), BOM Creation, Manufacturing Drawings, Fabrication Drawings, Engineering Change Management, Design Review

Standards & Codes: IS 2062, ASME B31.3 (reference), SS 304 / SS 316L / SS 409M material standards

Non-Destructive Testing: Ultrasonic Testing (UT), Magnetic Particle Testing (MPT), Penetrant Testing (PT), Radiographic Testing (RT), Real-Time Radiography (RTFI)

Soft Skills: Cross-functional Collaboration, Problem Solving, Design Reviews, Engineering Documentation, Continuous Learning

CERTIFICATIONS & TRAINING

- Certified SolidWorks Professional (CSWP) — Dassault Systèmes | December 16, 2025
- Non-Destructive Testing (NDT) — UT, MPT, PT, RT, RTFI | March 2019 – May 2021
- Electric Vehicle Technology Workshop — Skill-Lync | Fundamentals of BMS, electric motors, and power electronics
- Mechanical Component & Engine Testing Workshop — Mahindra & Mahindra | Hands-on automotive testing methodologies | June 2025

EDUCATION

Anand Institute of Higher Technology

2020 – 2024

Bachelor of Engineering — Mechanical Engineering | CGPA: 8 / 10

Government Higher Secondary School, Pazhampet

2020

Class XII — 69%

RC St. Joseph's High School, Mamallapuram

2018

Class X — 56%

LANGUAGES

English: Professional Working Proficiency

Tamil: Native